

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Marks available |     |     |       |       |      |
|----------|--|--|--|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|-----|-----|-------|-------|------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | AO1             | AO2 | AO3 | Total | Maths | Prac |
| 4        |  |  |  | <p>Indicative content:</p> <ul style="list-style-type: none"><li>• receptors in skin receive the stimulus</li><li>• impulse</li><li>• transmitted to coordinator/spinal cord</li><li>• along sensory neurone/nerve cell</li><li>• then to relay neurone/nerve cell</li><li>• then to motor neurone/nerve cell</li><li>• then to muscle/effector</li><li>• which contracts withdrawing hand</li><li>• one correct reference to synapses</li></ul> <p><b>5-6 marks</b><br/>At least seven points from indicative content<br/><i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i></p> <p><b>3-4 marks</b><br/>At least four points from indicative content<br/><i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i></p> | 6               |     |     |       |       |      |

| Question |  |  |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                          | Marks available |          |          |          |          |          |
|----------|--|--|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|----------|----------|----------|
|          |  |  |  |                                                                                                                                                                                                                                                                                                                                                                                                                          | AO1             | AO2      | AO3      | Total    | Maths    | Prac     |
|          |  |  |  | <p><b>1-2 marks</b><br/>At least one point from indicative content<br/><i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i></p> <p><b>0 marks</b><br/><i>No attempt made or no response worthy of credit.</i></p> |                 |          |          |          |          |          |
|          |  |  |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                                  | <b>6</b>        | <b>0</b> | <b>0</b> | <b>6</b> | <b>0</b> | <b>0</b> |